HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Engineered **Python** data pipelines on **AWS Lambda** to process ~3,500 events/sec, replacing a brittle scheduled-batch job for internal ML workflows.
- Designed and implemented **auto-scaling architectures** for online inference pipelines using **Kubernetes**, ensuring optimized performance during traffic spikes.
- Built and maintained inference engines using **vLLM** and **NVIDIA Triton**, delivering low-latency AI serving solutions for enterprise applications.
- Developed **Java Spring Boot** REST APIs for cross-service data exchange, incorporating request validation and circuit-breaker patterns.
- Profiled and optimized models for specialized hardware backends including **NVIDIA GPUs** and **Apple Silicon**, maximizing computational environment utilization.
- Established observability frameworks to monitor **Time to First Token (TTFT)** and tokens per second against service-level agreements.
- Containerized services with **Docker** and deployed them to a **Kubernetes cluster (EKS)**, configuring resources for predictable rollouts.

SKILLS

Programming Languages – Python, C++, Rust, Golang, Java

Inference Tools – vLLM, Text Generation Inference (TGI), NVIDIA Triton, TensorRT, Llama.cpp, Ollama

Infrastructure – Docker, Kubernetes, AWS, GCP, Azure, Jenkins, Git

Hardware Acceleration – NVIDIA GPUs, CUDA, TensorRT, Apple Silicon, CoreML, TPUs, LPUs

MLOps & Observability – MLOps, SRE, Observability Frameworks, Performance Testing, Load Testing, TTFT, Tokens per Second, Memory Bandwidth Utilization

PROJECTS

High-Performance AI Inference Service

- Developed a scalable inference service using **Python** and **FastAPI** to serve machine learning models, handling up to 1,500 requests per minute.
- Optimized model deployment for low-latency serving on **AWS EC2 GPU instances**, reducing average inference time by **30%**.
- Integrated **Docker** for containerization and managed deployments via **Kubernetes** for reliable scaling and orchestration.

LLM Deployment and Optimization

- Engineered a deployment pipeline for a transformer-based LLM using **vLLM** on **NVIDIA GPUs**, improving throughput by **40%**.
- Implemented **Kubernetes**-based auto-scaling for the LLM inference endpoints, ensuring stability during peak loads of 500 concurrent users.
- Configured **Prometheus and Grafana** for real-time monitoring of key performance indicators like latency and token generation rate.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)